2026Fall Quiz

for (MATH230) Probability & Statistics

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Problem 1.1. Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated

(a) with no restrictions?

(b) if each couple is to sit together?

(c) if at least one couple cannot sit next to each other?

(d) if all the men sit together to the right of all the women?

Note. You can write your answer in calculator-ready form. You don't have to calculate the exact value of fractions or combinations. For example, $8!$, $(4!)^4$, and $\binom{8}{3}/\binom{10}{3}$ are acceptable forms.

Solution.

(a) With no restrictions, the 8 people can be arranged in

$$\boxed{8!}$$

ways.

(b) Treat each couple as a single block. The four couple-blocks can be arranged in 4! ways, and the two people within each couple can be arranged in 2! ways. Hence the number of arrangements is

$$\boxed{4!(2!)^4}.$$

(c) The complement consists of the arrangements in which all four couples sit together. By parts (a) and (b), the required number is

$$\boxed{8! - 4!(2!)^4}.$$

(d) The four women must occupy the four leftmost seats and the four men the four rightmost seats. Each group can be arranged independently in 4! ways, giving

$$\boxed{4! \cdot 4!}.$$

□

Problem 1.2. A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.25. Of those that purchase latex paint, 75% also purchase rollers. But only 25% of semigloss paint buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?

Note. If you introduce symbols (e.g., A, B) in your solution, you must clearly define what each symbol represents.

Solution. Let

$$L = \{\text{the buyer purchases latex paint}\}, \qquad S = \{\text{the buyer purchases semigloss paint}\},$$

and

$$R = \{\text{the buyer purchases a roller}\}.$$

Then

$$\mathbb{P}(L) = \frac{1}{4}, \qquad \mathbb{P}(R \mid L) = \frac{3}{4}, \qquad \mathbb{P}(R \mid S) = \frac{1}{4},$$

and, since the paint is either latex or semigloss,

$$\mathbb{P}(S) = 1 - \mathbb{P}(L) = \frac{3}{4}.$$

By the law of total probability,

$$\begin{aligned} \mathbb{P}(R) &= \mathbb{P}(R \mid L)\mathbb{P}(L) + \mathbb{P}(R \mid S)\mathbb{P}(S) \\ &= \frac{3}{4} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{8}. \end{aligned}$$

Therefore, by Bayes' rule,

$$\boxed{\mathbb{P}(L \mid R) = \frac{\mathbb{P}(R \mid L)\mathbb{P}(L)}{\mathbb{P}(R)} = \frac{(3/4)(1/4)}{3/8} = \frac{1}{2}}.$$

Thus the probability that the paint is latex is 50%.

□

Problem 2.1. For a laboratory assignment, if the equipment is working, the density function of the observed outcome X is

$$f_X(x) = \begin{cases} c(2-x), & 0 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Find the value of c so that f_X is a valid density function.

(b) Find $\mathbb{P}(X < 1)$.

(c) Given that $X \geq 1$, what is the probability that X will be less than 1.5?

Solution.

(a) A probability density must integrate to 1. Thus

$$1 = \int_{-\infty}^{\infty} f_X(x)\ \mathrm{d}x = \int_0^2 c(2-x)\ \mathrm{d}x = 2c,$$

so

$$\boxed{c = \frac{1}{2}}.$$

(b) Using $c = 1/2$,

$$\boxed{\mathbb{P}(X < 1) = \int_0^1 \frac{1}{2}(2-x)\ \mathrm{d}x = \frac{3}{4}}.$$

(c) By the definition of conditional probability,

$$\mathbb{P}(X < 1.5 \mid X \geq 1) = \frac{\mathbb{P}(1 \leq X < 1.5)}{\mathbb{P}(X \geq 1)}.$$

Here,

$$\mathbb{P}(1 \leq X < 1.5) = \int_1^{3/2} \frac{1}{2}(2-x)\ \mathrm{d}x = \frac{3}{16},$$

while

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X < 1) = \frac{1}{4}.$$

Therefore,

$$\boxed{\mathbb{P}(X < 1.5 \mid X \geq 1) = \frac{3/16}{1/4} = \frac{3}{4}}.$$

□

Problem 2.2. The amount of kerosene, in thousands of liters, in a tank at the beginning of any day is a random amount Y from which a random amount X is sold during that day. Suppose that the tank is not resupplied during the day so that $x \leq y$, and assume that the joint density function of these variables is

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 < x \leq y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Find the marginal density function of X.

(b) Find the marginal density function of Y.

(c) Determine if X and Y are independent.

(d) Find $\mathbb{P}(1/2 < Y < 3/4 \mid X = 1/4)$.

Solution.

(a) For fixed x with $0 < x < 1$, the support has $x \leq y < 1$. Hence

$$f_X(x) = \int_x^1 f_{X,Y}(x,y)\ \mathrm{d}y = \int_x^1 2\ \mathrm{d}y = 2(1-x).$$

Therefore,

$$\boxed{f_X(x) = \begin{cases} 2(1-x), & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}}$$

(b) For fixed y with $0 < y < 1$, the support has $0 < x \leq y$. Thus

$$f_Y(y) = \int_0^y f_{X,Y}(x,y)\ \mathrm{d}x = \int_0^y 2\ \mathrm{d}x = 2y.$$

Therefore,

$$\boxed{f_Y(y) = \begin{cases} 2y, & 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}}$$

(c) If X and Y were independent, then $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ almost everywhere. But for $0 < y < x < 1$,

$$f_{X,Y}(x,y) = 0, \qquad f_X(x)f_Y(y) = 4y(1-x) > 0.$$

Hence

$$\boxed{X \text{ and } Y \text{ are not independent.}}$$

(d) For $0 < x < 1$, the conditional density of Y given $X = x$ is

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{1}{1-x}, \qquad x < y < 1.$$

When $X = 1/4$,

$$f_{Y|X}(y \mid 1/4) = \frac{4}{3}, \qquad \frac{1}{4} < y < 1.$$

Therefore,

$$\boxed{\mathbb{P}\left(\frac{1}{2} < Y < \frac{3}{4} \,\middle|\, X = \frac{1}{4}\right) = \int_{1/2}^{3/4} \frac{4}{3}\ \mathrm{d}y = \frac{1}{3}}.$$

□

Problem 3.1. The probability distribution of the discrete random variable X is

$$p_X(x) = \binom{5}{x}\left(\frac{1}{5}\right)^x\left(\frac{4}{5}\right)^{5-x}, \qquad x = 0, 1, 2, 3, 4, 5.$$

Find the mean of X.

Solution. By the definition of expectation for a discrete random variable,

$$\begin{aligned}\mathbb{E}[X] &= \sum_{x=0}^{5} x\, p_X(x) \\ &= \sum_{x=0}^{5} x\binom{5}{x}\left(\frac{1}{5}\right)^x\left(\frac{4}{5}\right)^{5-x} \\ &= \frac{1280 + 1280 + 480 + 80 + 5}{3125} \\ &= \boxed{1}.\end{aligned}$$

□

Problem 3.2. Let X and Y be random variables with joint density function

$$f_{X,Y}(x, y) = \begin{cases} 4xy, & 0 < x, y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Find the expected value of $X + Y$.

(b) Find the expected value of XY.

Solution.

(a) By the definition of expectation for a function of jointly continuous random variables,

$$\begin{aligned}\mathbb{E}[X+Y] &= \int_0^1\int_0^1 (x+y) f_{X,Y}(x,y)\ \mathrm{d}y\ \mathrm{d}x \\ &= \int_0^1\int_0^1 (4x^2y + 4xy^2)\ \mathrm{d}y\ \mathrm{d}x \\ &= \int_0^1 \left(2x^2 + \frac{4}{3}x\right)\ \mathrm{d}x \\ &= \boxed{\frac{4}{3}}.\end{aligned}$$

(b) Again using the definition of expectation,

$$\begin{aligned}\mathbb{E}[XY] &= \int_0^1\int_0^1 xy\, f_{X,Y}(x,y)\ \mathrm{d}y\ \mathrm{d}x \\ &= 4\int_0^1\int_0^1 x^2y^2\ \mathrm{d}y\ \mathrm{d}x \\ &= 4\left(\int_0^1 x^2\ \mathrm{d}x\right)\left(\int_0^1 y^2\ \mathrm{d}y\right) \\ &= \boxed{\frac{4}{9}}.\end{aligned}$$

□

Problem 4.1. From a sack of fruit containing 2 oranges, 1 apple, and 2 bananas, a random sample of 3 pieces of fruit is selected. If X is the number of oranges and Y is the number of apples in the sample, find the covariance of the random variables X and Y.

Solution. There are $\binom{5}{3} = 10$ equally likely samples. To obtain $X = x$ oranges and $Y = y$ apples, the sample must contain $3 - x - y$ bananas. Hence the joint probability distribution is

$$p_{X,Y}(x,y) = \frac{\binom{2}{x}\binom{1}{y}\binom{2}{3-x-y}}{\binom{5}{3}}$$

for the admissible values of x and y, giving

$p_{X,Y}(x,y)$	x		
y	0	1	2
0	0	$\frac{2}{10}$	$\frac{2}{10}$
1	$\frac{1}{10}$	$\frac{4}{10}$	$\frac{1}{10}$

The marginal probabilities give

$$\mathbb{E}[X] = (1)\left(\frac{2+4}{10}\right) + (2)\left(\frac{2+1}{10}\right) = \frac{6}{5},$$
$$\mathbb{E}[Y] = (1)\left(\frac{1+4+1}{10}\right) = \frac{3}{5}.$$

Also, only the pairs $(1,1)$ and $(2,1)$ contribute to $\mathbb{E}[XY]$:

$$\begin{aligned}\mathbb{E}[XY] &= \sum_x \sum_y xy\, p_{X,Y}(x,y) \\ &= (1)(1)\frac{4}{10} + (2)(1)\frac{1}{10} = \frac{3}{5}.\end{aligned}$$

Therefore, by the covariance formula,

$$\boxed{\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \frac{3}{5} - \left(\frac{6}{5}\right)\left(\frac{3}{5}\right) = -\frac{3}{25}}.$$

□

Problem 4.2. Let X and Y be random variables with joint density function

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{1}{2}, & 0 < x \le y < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Find $\mathrm{Var}(X)$ and $\mathrm{Var}(Y)$.

(b) Find $\mathrm{Cov}(X,Y)$.

(c) Find $\rho_{X,Y}$, the correlation coefficient between X and Y.

(d) Find $\mathrm{Var}(X+Y)$.

Hint: You may use $\mathbb{E}[X] = \frac{2}{3}$, $\mathbb{E}[Y] = \frac{4}{3}$, $\mathbb{E}[X^2] = \frac{2}{3}$, and $\mathbb{E}[Y^2] = 2$.

Solution. The support is the triangular region $0 < x \le y < 2$.

(a) Using the given moments and the variance formula,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{2}{3} - \left(\frac{2}{3}\right)^2 = \boxed{\frac{2}{9}},$$

$$\mathrm{Var}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = 2 - \left(\frac{4}{3}\right)^2 = \boxed{\frac{2}{9}}.$$

(b) We first calculate the mixed moment using the joint density:

$$\begin{aligned} \mathbb{E}[XY] &= \int_0^2 \int_0^y xy\, f_{X,Y}(x,y)\, \mathrm{d}x\, \mathrm{d}y \\ &= \int_0^2 \int_0^y \frac{xy}{2}\, \mathrm{d}x\, \mathrm{d}y \\ &= \frac{1}{4}\int_0^2 y^3\, \mathrm{d}y = 1. \end{aligned}$$

Therefore,

$$\boxed{\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 1 - \left(\frac{2}{3}\right)\left(\frac{4}{3}\right) = \frac{1}{9}}.$$

(c) Since both variances are positive, the correlation coefficient is

$$\boxed{\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}} = \frac{1/9}{\sqrt{(2/9)(2/9)}} = \frac{1}{2}}.$$

(d) Using the variance of a sum,

$$\begin{aligned} \mathrm{Var}(X+Y) &= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y) \\ &= \frac{2}{9} + \frac{2}{9} + 2\left(\frac{1}{9}\right) = \boxed{\frac{2}{3}}. \end{aligned}$$

□

Problem 5.1. Let X be a random variable with mean $\mu = 12$ and variance $\sigma^2 = 9$. Using Chebyshev's inequality, answer the following questions.

(a) Find an upper bound for $\mathbb{P}(|X - 12| \ge 6)$.

(b) Find a lower bound for $\mathbb{P}(3 < X < 21)$.

(c) Find the smallest positive constant c for which Chebyshev's inequality guarantees that

$$\mathbb{P}(|X - 12| < c) \ge \frac{15}{16}.$$

Solution. Since $\mathbb{E}[X] = 12$ and $\mathrm{Var}(X) = 9$, Chebyshev's inequality gives, for every $t > 0$,

$$\mathbb{P}(|X - 12| \geq t) \leq \frac{9}{t^2}, \qquad \mathbb{P}(|X - 12| < t) \geq 1 - \frac{9}{t^2}.$$

(a) Setting $t = 6$ yields

$$\boxed{\mathbb{P}(|X - 12| \geq 6) \leq \frac{9}{6^2} = \frac{1}{4}}.$$

(b) Since $3 < X < 21$ is equivalent to $|X - 12| < 9$,

$$\boxed{\mathbb{P}(3 < X < 21) \geq 1 - \frac{9}{9^2} = \frac{8}{9}}.$$

(c) The guaranteed lower bound must be at least $15/16$:

$$1 - \frac{9}{c^2} \geq \frac{15}{16} \iff \frac{9}{c^2} \leq \frac{1}{16} \iff c^2 \geq 144.$$

For $c > 0$, the smallest permitted constant is $\boxed{c = 12}$.

□

Problem 5.2. According to a study, approximately 30% of pipework failures in a chemical plant are due to operator error. Assume that the causes of successive failures are independent and that each failure is due to operator error with probability 0.30. Let X denote the number of operator-error failures among the next 8 failures. Use the binomial probability table below to answer the questions.

Table A.1 (continued). Binomial probability sums $B(r; n, p) = \sum_{x=0}^{r} \binom{n}{x} p^x (1-p)^{n-x}$.

		p									
n	r	0.10	0.20	0.25	0.30	0.40	0.50	0.60	0.70	0.80	0.90
8	0	0.4305	0.1678	0.1001	0.0576	0.0168	0.0039	0.0007	0.0001	0.0000	
	1	0.8131	0.5033	0.3671	0.2553	0.1064	0.0352	0.0085	0.0013	0.0001	
	2	0.9619	0.7969	0.6785	0.5518	0.3154	0.1445	0.0498	0.0113	0.0012	0.0000
	3	0.9950	0.9437	0.8862	0.8059	0.5941	0.3633	0.1737	0.0580	0.0104	0.0004
	4	0.9996	0.9896	0.9727	0.9420	0.8263	0.6367	0.4059	0.1941	0.0563	0.0050
	5	1.0000	0.9988	0.9958	0.9887	0.9502	0.8555	0.6846	0.4482	0.2031	0.0381
	6		0.9999	0.9996	0.9987	0.9915	0.9648	0.8936	0.7447	0.4967	0.1869
	7		1.0000	1.0000	0.9999	0.9993	0.9961	0.9832	0.9424	0.8322	0.5695
	8				1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

(a) State the distribution of X and write its probability mass function.

(b) What is the probability that at most 2 of the next 8 failures are due to operator error?

(c) What is the probability that at least 4 of the next 8 failures are due to operator error?

(d) Find $\mathbb{P}(2 \leq X \leq 5)$.

Solution.

(a) There are 8 independent Bernoulli trials, each with operator-error probability 0.30. Therefore,

$$\boxed{X \sim \mathrm{Bin}(8, 0.30)}, \qquad \boxed{\mathbb{P}(X = x) = \binom{8}{x} (0.30)^x (0.70)^{8-x}}, \quad x = 0, 1, \ldots, 8.$$

(b) The table gives the cumulative probability for $n = 8$, $p = 0.30$, and $r = 2$:

$$\boxed{\mathbb{P}(X \leq 2) = B(2; 8, 0.30) = 0.5518}.$$

(c) By the complement rule and the table entry at $r = 3$,

$$\boxed{\mathbb{P}(X \geq 4) = 1 - \mathbb{P}(X \leq 3) = 1 - 0.8059 = 0.1941}.$$

(d) Subtract cumulative probabilities at $r = 5$ and $r = 1$:

$$\boxed{\mathbb{P}(2 \leq X \leq 5) = \mathbb{P}(X \leq 5) - \mathbb{P}(X \leq 1) = 0.9887 - 0.2553 = 0.7334}.$$

□

Problem 6.1. Hospital administrators must plan for patient traffic in an emergency room. At a certain hospital, the staff cannot accommodate the patient traffic if more than 3 emergency cases arrive in a given hour. Assume emergency cases arrive according to a Poisson process at an average rate of 1.5 cases per hour.

(a) What is the probability that the staff cannot accommodate the patient traffic in a given hour? (25 points)

(b) What is the probability that exactly 5 emergency cases arrive during a 3-hour shift? (25 points)

The following table gives the Poisson cumulative probabilities $\mathbb{P}(N \leq r) = \sum_{x=0}^{r} e^{-\mu}\mu^x/x!$.

$r\backslash\mu$	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
0	0.3679	0.2231	0.1353	0.0821	0.0498	0.0302	0.0183	0.0111	0.0067
1	0.7358	0.5578	0.4060	0.2873	0.1991	0.1359	0.0916	0.0611	0.0404
2	0.9197	0.8088	0.6767	0.5438	0.4232	0.3208	0.2381	0.1736	0.1247
3	0.9810	0.9344	0.8571	0.7576	0.6472	0.5366	0.4335	0.3423	0.2650
4	0.9963	0.9814	0.9473	0.8912	0.8153	0.7254	0.6288	0.5321	0.4405
5	0.9994	0.9955	0.9834	0.9580	0.9161	0.8576	0.7851	0.7029	0.6160

Solution. Let N_t count emergency arrivals during t hours. Since the rate is 1.5 per hour, $N_t \sim \text{Pois}(1.5t)$.

(a) The one-hour count satisfies $N_1 \sim \text{Pois}(1.5)$. The staff cannot accommodate the traffic when $N_1 > 3$. From row $r = 3$, column $\mu = 1.5$,

$$\boxed{\mathbb{P}(N_1 > 3) = 1 - \mathbb{P}(N_1 \leq 3) = 1 - 0.9344 = 0.0656}.$$

(b) In three hours the mean count is $1.5(3) = 4.5$. From row $r = 5$ and row $r = 4$ at $\mu = 4.5$,

$$\boxed{\mathbb{P}(N_3 = 5) = \mathbb{P}(N_3 \leq 5) - \mathbb{P}(N_3 \leq 4) = 0.7029 - 0.5321 = 0.1708}.$$

□

Problem 6.2. Let X have a normal distribution with mean $\mu = 40$ and standard deviation $\sigma = 5$. Find

(a) the area under the normal curve to the right of $x = 35$. (10 points)

(b) the area under the normal curve between $x = 42$ and $x = 47$. (15 points)

(c) the value of x that leaves 80% of the area to its left. (10 points)

(d) the two values of x that contain the middle 80% of the area. (15 points)

The following table gives $\Phi(z) = \mathbb{P}(Z \le z)$ for $Z \sim \mathcal{N}(0,1)$, with z equal to the row value plus the column value.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319

Solution. Standardize with $Z = (X - 40)/5 \sim \mathcal{N}(0,1)$.

(a) Since $z = (35 - 40)/5 = -1$ and the normal curve is symmetric,

$$\boxed{\mathbb{P}(X > 35) = 1 - \Phi(-1) = \Phi(1) = 0.8413}.$$

(b) The standardized endpoints are $z_1 = (42 - 40)/5 = 0.4$ and $z_2 = (47 - 40)/5 = 1.4$. Thus

$$\boxed{\mathbb{P}(42 < X < 47) = \Phi(1.4) - \Phi(0.4) = 0.9192 - 0.6554 = 0.2638}.$$

(c) The required standardized percentile satisfies $\Phi(z) = 0.80$, giving $z \approx 0.84$ from the table. Hence

$$\boxed{x = 40 + 5(0.84) \approx 44.2}.$$

(d) The middle 80% leaves 10% in each tail. The upper standardized boundary is $z_{0.90} \approx 1.28$ (and the lower is -1.28). Therefore

$$\boxed{x_L = 40 - 5(1.28) \approx 33.6}, \qquad \boxed{x_U = 40 + 5(1.28) \approx 46.4}.$$

□